Analyzing the Impact of Sleep and Lifestyle Factors on Stress Levels using ML

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Introduction:

Sleep plays a critical role in maintaining physical and mental well-being. In modern society, increasing work demands, sedentary lifestyles, and stress have significantly affected sleep patterns and overall health. Poor sleep quality is often associated with increased stress levels, reduced physical activity, obesity, cardiovascular issues, and other health complications.

Machine Learning techniques can be used to identify patterns in healthcare datasets and predict important health outcomes. By analyzing relationships among sleep duration, sleep quality, physical activity, body mass index (BMI), heart rate, blood pressure, and stress levels, predictive models can assist in understanding the factors that influence an individual's health.

This project aims to analyze the Sleep Health and Lifestyle Dataset and develop machine learning models capable of predicting stress levels using sleep-related and lifestyle factors. The study also explores the relationships among various health indicators through data visualization and statistical analysis.

Problem Statement:

Predictive Modeling of Sleep Quality and Stress Architecture: Assessing Interdependencies between Lifestyle, BMI, and Cardiovascular Metrics.
The objective of this project is to investigate how lifestyle factors and physiological measurements influence stress levels. The project utilizes machine learning techniques to identify significant predictors of stress and evaluate their relationships with sleep quality and cardiovascular health.

Objectives:

1. To analyze sleep and lifestyle-related health data.
2. To identify relationships between sleep duration and stress levels.
3. To examine the influence of BMI, physical activity, heart rate, and blood pressure on stress.
4. To visualize healthcare data using graphical techniques.
5. To build machine learning models for stress level prediction.
6. To compare model performance and identify the most effective prediction algorithm.

Dataset Description

The dataset used in this project is the Sleep Health and Lifestyle Dataset obtained from Kaggle. The dataset contains information related to sleep patterns, lifestyle factors, cardiovascular health indicators, and stress levels of individuals.

Dataset Link: https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset

The dataset contains information related to sleep patterns, lifestyle factors, and cardiovascular health indicators of individuals.

Dataset Characteristics:

- Number of Records: 374
- Number of Attributes: 13
- Dataset Type: Healthcare and Lifestyle Dataset

The dataset contains the following attributes:

1. Person ID
2. Gender
3. Age
4. Occupation
5. Sleep Duration
6. Quality of Sleep
7. Physical Activity Level
8. Stress Level
9. BMI Category
10. Blood Pressure
11. Heart Rate
12. Daily Steps
13. Sleep Disorder

Target Variable: Stress Level

Predictor Variables:

- Age
- Gender
- Occupation
- Sleep Duration
- Quality of Sleep
- Physical Activity Level
- BMI Category
- Blood Pressure
- Heart Rate
- Daily Steps
- Sleep Disorder

Initial Data Exploration:

The dataset was imported into Google Colab using the Pandas library. Preliminary analysis was performed to understand the dataset structure, identify missing values, and examine variable types.

Data Type Analysis

The dataset consists of:

- Integer variables: 7
- Floating-point variables: 1
- Categorical variables: 5

The presence of both numerical and categorical features makes the dataset suitable for machine learning classification tasks.

Missing Value Analysis:

Missing value analysis was performed using the isnull() function.

Observation:

All attributes except sleep disorder contained complete information. The Sleep Disorder attribute contained 219 missing values. These missing values likely indicate the absence of diagnosed sleep disorders for several individuals.

Descriptive Statistics:

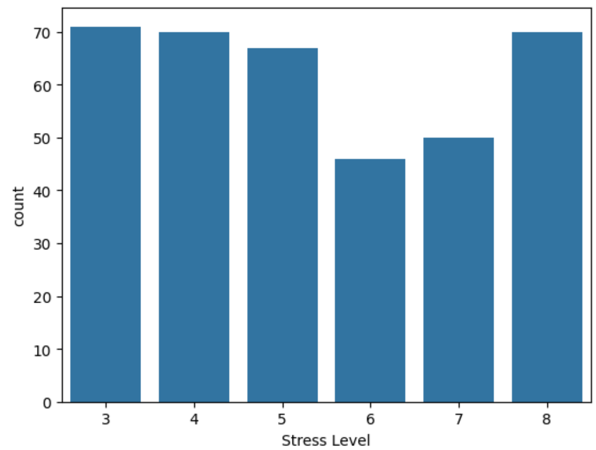
Key observations from descriptive analysis include:

- Average Age: 42.18 years
- Average Sleep Duration: 7.13 hours
- Average Stress Level: 5.39
- Average Heart Rate: 70.17 beats per minute
- Average Daily Steps: 6816.84

These values indicate that the dataset primarily represents middle-aged adults with moderate sleep duration and moderate stress levels.

Stress Level Distribution:

A count plot was generated to examine the distribution of stress levels across participants.

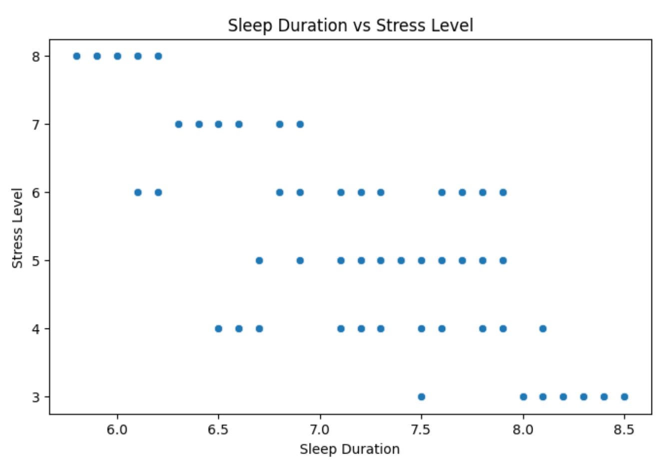


Observation:

Stress levels 3, 4, and 8 were among the most frequently observed categories, while stress level 6 appeared less frequently. The distribution is reasonably balanced, making the dataset appropriate for classification-based machine learning models.

Relationship between Sleep Duration and Stress Level:

A scatter plot was generated to investigate the relationship between sleep duration and stress level.

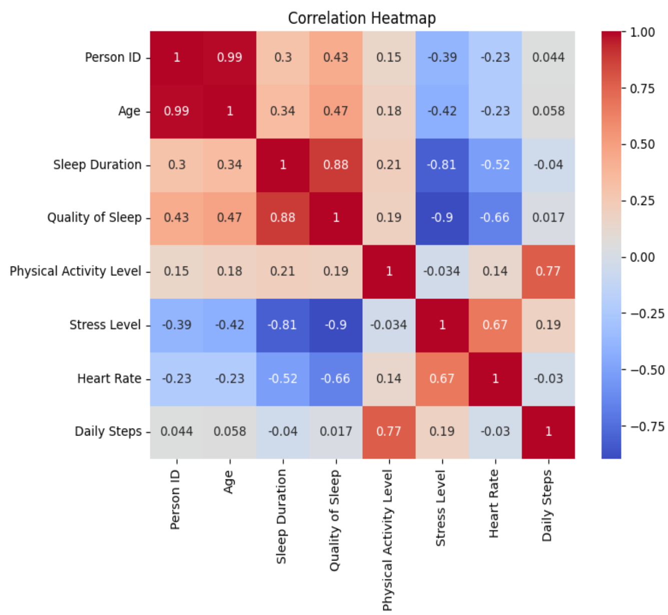


Observation:

A negative relationship was observed between sleep duration and stress level. Individuals with shorter sleep durations generally exhibited higher stress levels, whereas individuals with longer sleep durations tended to report lower stress levels. This observation supports existing research suggesting that insufficient sleep contributes to increased psychological stress.

EXPLORATORY DATA ANALYSIS:

Figure 1: Correlation Heatmap



Objective:

The purpose of correlation analysis is to identify relationships among the variables in the dataset and determine which factors are most strongly associated with stress levels. Correlation coefficients range from -1 to +1, where values closer to +1 indicate a strong positive relationship, values closer to -1 indicate a strong negative relationship, and values near 0 indicate little or no relationship.

Observation:

The correlation heatmap reveals several significant relationships among the variables:

- Quality of Sleep exhibits a strong negative correlation with Stress Level (-0.90).

- Sleep Duration shows a strong negative correlation with Stress Level (-0.81).
- Heart Rate has a strong positive correlation with Stress Level (0.67).
- Age demonstrates a moderate negative correlation with Stress Level (-0.42).
- Physical Activity Level has a negligible correlation with Stress Level (-0.03).
- Daily Steps show a weak positive correlation with Stress Level (0.19).

Additionally:

- Sleep Duration and Quality of Sleep are strongly positively correlated (0.88).
- Physical Activity Level and Daily Steps show a strong positive relationship (0.77).
- Quality of Sleep and Heart Rate exhibit a moderate negative correlation (-0.66).

Interpretation:

The results indicate that sleep-related factors are the most influential variables affecting stress levels within the dataset. Individuals who sleep longer and experience better sleep quality tend to have significantly lower stress levels.

The strong positive correlation between Heart Rate and Stress Level suggests that increased stress may lead to elevated heart rates, making heart rate a useful physiological indicator of stress.

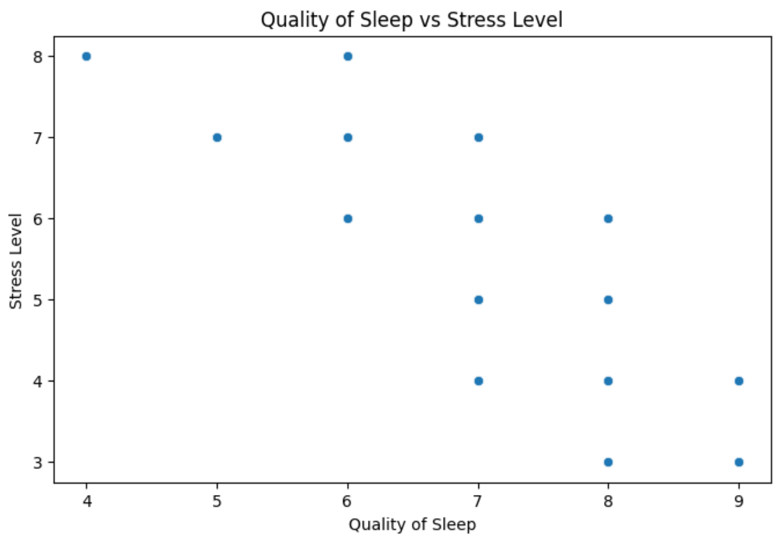
The moderate negative correlation between Age and Stress Level indicates that older individuals in the dataset generally reported slightly lower stress levels.

The very weak relationship between Physical Activity Level and Stress Level suggests that physical activity alone may not directly influence stress levels in this dataset, although it may contribute indirectly through improved sleep quality and overall health.

Conclusion

Based on the correlation analysis, Quality of Sleep, Sleep Duration, and Heart Rate emerged as the most significant predictors of Stress Level. Among these variables, Quality of Sleep demonstrated the strongest relationship with stress, followed by Sleep Duration and Heart Rate. These findings suggest that sleep-related factors play a critical role in determining stress levels and should be considered primary features for subsequent machine learning model development.

Figure 2: Quality of sleep vs stress level



Objective:

The objective of this visualization is to examine the relationship between sleep quality and stress levels among individuals.

Observation:

The scatter plot shows a downward trend between Quality of Sleep and Stress Level. Higher sleep quality scores are generally associated with lower stress levels. Individuals with sleep quality ratings between 8 and 9 tend to exhibit stress levels between 3 and 5, whereas individuals with lower sleep quality ratings tend to report higher stress levels.

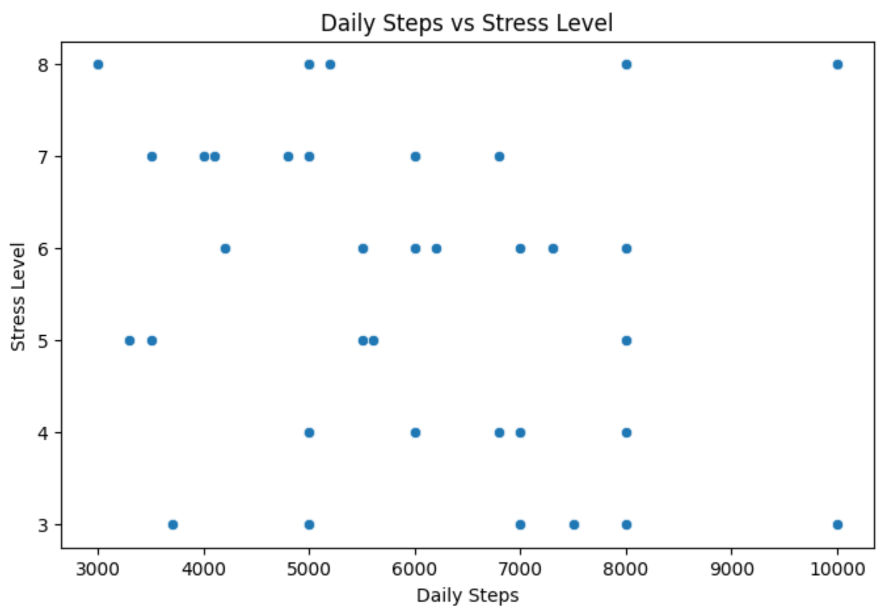
Interpretation:

The observed pattern indicates a strong negative relationship between sleep quality and stress. As sleep quality improves, stress levels tend to decrease. This finding supports the hypothesis that better sleep quality contributes to improved mental well-being.

Conclusion:

Sleep quality appears to be one of the strongest factors influencing stress levels and may serve as an important predictor in stress prediction models.

Figure 3: Daily steps vs Stress level



Objective:

The objective of this analysis is to investigate whether daily physical activity, represented by daily step count, influences stress levels.

Observation:

The scatter plot does not exhibit a clear upward or downward trend. Individuals with both low and high daily step counts display varying stress levels. Stress levels are distributed across the entire range of daily steps.

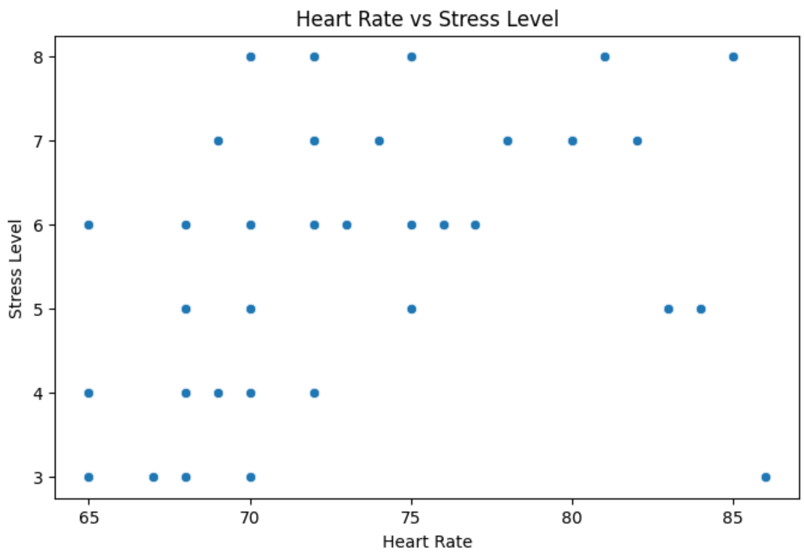
Interpretation:

The absence of a clear pattern suggests that daily step count alone may not significantly affect stress levels. Other factors such as sleep quality, sleep duration, and physiological indicators may have a stronger influence on stress.

Conclusion:

Daily steps show only a weak association with stress level and may not be a primary predictor for stress prediction.

Figure 4: Heart rate vs Stress level



Objective:

The purpose of this analysis is to examine the relationship between heart rate and stress level.

Observation:

The scatter plot indicates that higher stress levels are generally associated with increased heart rates. Individuals with stress levels of 7 and 8 frequently exhibit heart rates above 75 beats per minute.

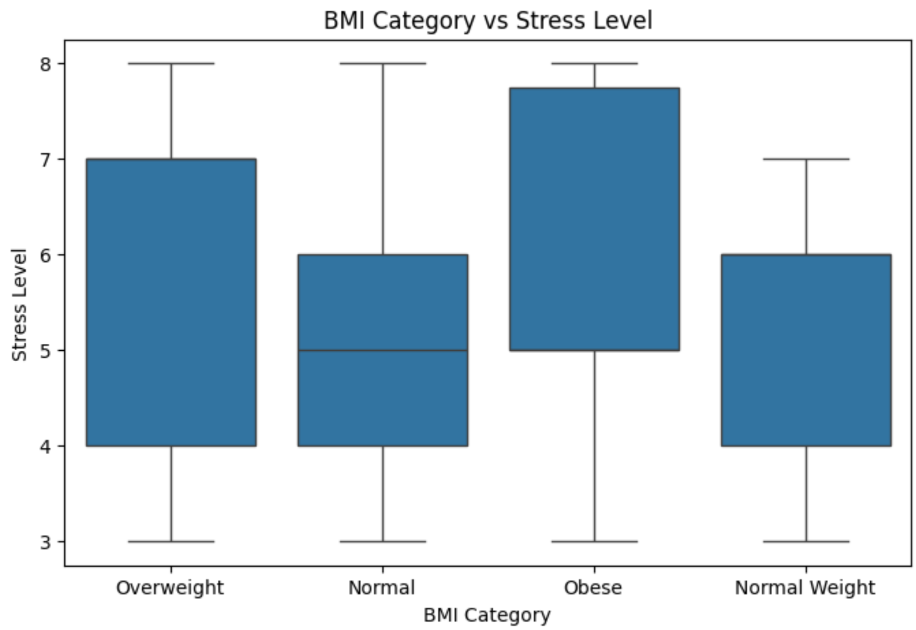
Interpretation:

This pattern suggests a positive relationship between heart rate and stress. Elevated stress may trigger physiological responses that increase heart rate.

Conclusion:

Heart rate is a meaningful physiological indicator of stress and can be considered an important feature for stress prediction.

Figure 5: BMI Category vs Stress level



Objective:

The objective of this analysis is to compare stress levels across different BMI categories.

Observation:

The box plot reveals that stress levels vary across BMI groups. The Obese category exhibits relatively higher median stress levels and a wider spread compared to other categories. Normal Weight individuals generally display lower and more consistent stress levels.

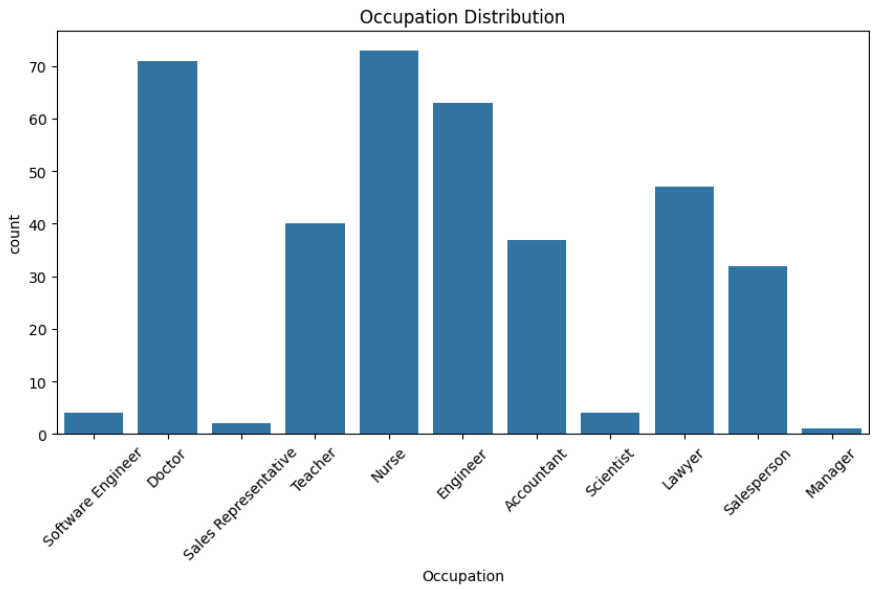
Interpretation:

The results suggest that body weight status may influence stress levels. Higher BMI categories appear to be associated with elevated stress, although variability exists within each group.

Conclusion:

BMI category may contribute to stress prediction; however, its influence appears weaker compared to sleep-related variables.

Figure 6: Occupation Distribution



Objective:

The objective of this visualization is to understand the composition of occupations represented in the dataset.

Observation:

The dataset contains individuals from multiple occupations. Nurses, Doctors, and Engineers constitute the largest proportion of participants, while Managers, Scientists, and Sales Representatives are comparatively less represented.

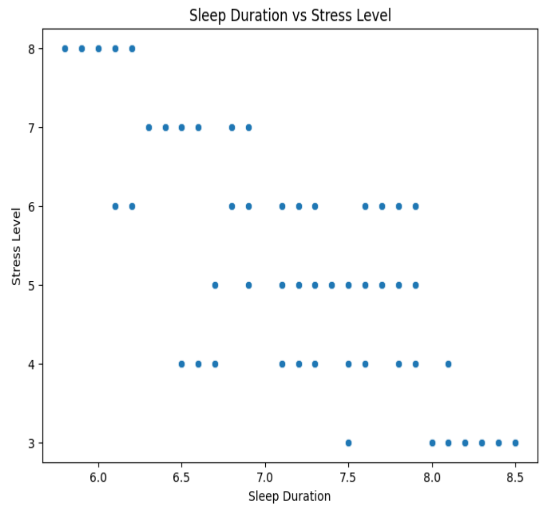
Interpretation:

The dataset provides occupational diversity, allowing analysis of stress patterns across different professions. However, the imbalance in occupation frequencies should be considered when interpreting results.

Conclusion:

The dataset includes a broad range of occupations, providing valuable demographic information for stress analysis and prediction.

Figure 7: Sleep Duration vs Stress Level



Objective

The objective of this visualization is to examine the relationship between sleep duration and stress level, and to determine whether the amount of sleep influences an individual's stress level.

Observation

The scatter plot shows a clear negative relationship between Sleep Duration and Stress Level. Individuals with shorter sleep durations (approximately 5.8–6.5 hours) generally exhibit higher stress levels (6–8), whereas individuals who sleep for longer durations (approximately 7.8–8.5 hours) tend to have lower stress levels (3–4). The data points indicate that stress decreases as sleep duration increases.

Interpretation

The graph suggests that adequate sleep plays an important role in stress management. Individuals who obtain sufficient sleep are generally associated with lower stress levels, while reduced sleep duration is associated with increased stress. This observation is further supported by the correlation analysis, which showed a strong negative correlation (-0.81) between Sleep Duration and Stress Level. Furthermore, Random Forest feature importance analysis identified Sleep Duration as the most influential predictor of stress level, indicating its significant contribution to the machine learning model.

Conclusion

The analysis indicates that Sleep Duration is a major factor influencing stress levels. Maintaining sufficient sleep duration may help reduce stress and improve overall well-being. The results also demonstrate that Sleep Duration is the most important feature for predicting stress levels in the developed machine learning model.

From all the graphs attached, we can conclude that

1. Sleep Quality demonstrated the strongest negative relationship with Stress Level.
2. Sleep Duration was also strongly associated with reduced stress.
3. Heart Rate showed a significant positive relationship with Stress Level.
4. Daily Steps exhibited only a weak relationship with stress.
5. BMI categories showed moderate variation in stress levels.
6. The dataset contained diverse occupational groups, enhancing the representativeness of the analysis.
7. Sleep-related variables emerged as the most influential factors for stress prediction.

Data Preprocessing

Before model development, the dataset was preprocessed to improve its suitability for machine learning algorithms.

Removal of Irrelevant Attributes

The "Person ID" attribute was removed from the dataset because it serves only as a unique identifier and does not provide meaningful information for stress level prediction. Retaining such identifiers may introduce noise and reduce model generalization.

Handling Missing Values

Missing value analysis revealed that only the "Sleep Disorder" attribute contained missing entries. Since the missing values likely represented individuals without diagnosed sleep disorders, the remaining variables were retained for analysis.

Categorical Variable Encoding

Several attributes, including Gender, Occupation, BMI Category, Blood Pressure, and Sleep Disorder, were categorical in nature. Machine learning algorithms require numerical inputs; therefore, Label Encoding was applied to convert categorical values into numerical representations.

Feature and Target Selection

The Stress Level attribute was selected as the target variable because the primary objective of the study was to predict an individual's stress category.

Target Variable:

- Stress Level

Predictor Variables:

- Gender
- Age
- Occupation
- Sleep Duration
- Quality of Sleep
- Physical Activity Level
- BMI Category
- Blood Pressure
- Heart Rate
- Daily Steps
- Sleep Disorder

Train-Test Split

The dataset was divided into training and testing subsets using an 80:20 ratio. Approximately 80% of the data was used to train the machine learning models, while the remaining 20% was reserved for evaluating model performance on unseen data.

Machine Learning Model Development

- Three classification algorithms were implemented and evaluated to predict stress levels based on sleep, lifestyle, and cardiovascular indicators.

1.Logistic Regression:

- Logistic Regression was used as a baseline classification model due to its simplicity and computational efficiency. The model attempts to identify relationships between predictor variables and stress levels using a probabilistic approach.

Result:

- Accuracy: 86.67%

Interpretation:

- The model demonstrated satisfactory predictive performance. However, because stress is influenced by multiple interacting variables, the linear assumptions of Logistic Regression limited its ability to capture complex relationships within the dataset.

2.Decision Tree Classifier:

- The Decision Tree algorithm was employed to model nonlinear relationships among the predictor variables. The model generates a hierarchical structure of decision rules based on feature values.

Result:

- Accuracy: 97.33%

Interpretation:

- The Decision Tree achieved significantly higher accuracy than Logistic Regression. This suggests that stress levels are influenced by nonlinear interactions among sleep quality, sleep duration, cardiovascular indicators, and lifestyle factors.

3.Random Forest Classifier

- Random Forest is an ensemble learning algorithm that combines multiple decision trees and aggregates their predictions through a voting mechanism. This approach improves robustness and reduces overfitting.

Result:

- Accuracy: 100.00%

Interpretation:

- Random Forest achieved the highest predictive performance among all evaluated models. The ensemble structure successfully captured complex patterns present in the healthcare dataset and produced highly accurate predictions.

Model Performance Comparison

The predictive performance of the three machine learning models was evaluated using classification accuracy.

Model	**Accuracy (%)**
Logistic Regression	86.67
Decision Tree	97.33
Random Forest	100.00

Analysis of Results:

The results indicate a progressive improvement in predictive performance as model complexity increases.

Logistic Regression provided a strong baseline performance but was unable to fully capture nonlinear relationships present in the dataset. Decision Tree significantly improved prediction accuracy by modeling complex decision boundaries. Random Forest further enhanced performance by combining multiple decision trees, thereby improving prediction stability and reducing classification errors.

The exceptionally high performance of the Random Forest model suggests that the dataset contains strong predictive signals, particularly from sleep-related variables such as Quality of Sleep and Sleep Duration.

Key Findings:

1. Quality of Sleep emerged as the strongest predictor of Stress Level.
2. Sleep Duration demonstrated a strong inverse relationship with Stress Level.
3. Heart Rate showed a meaningful positive association with Stress Level.

4. Random Forest achieved the best overall performance for stress prediction.

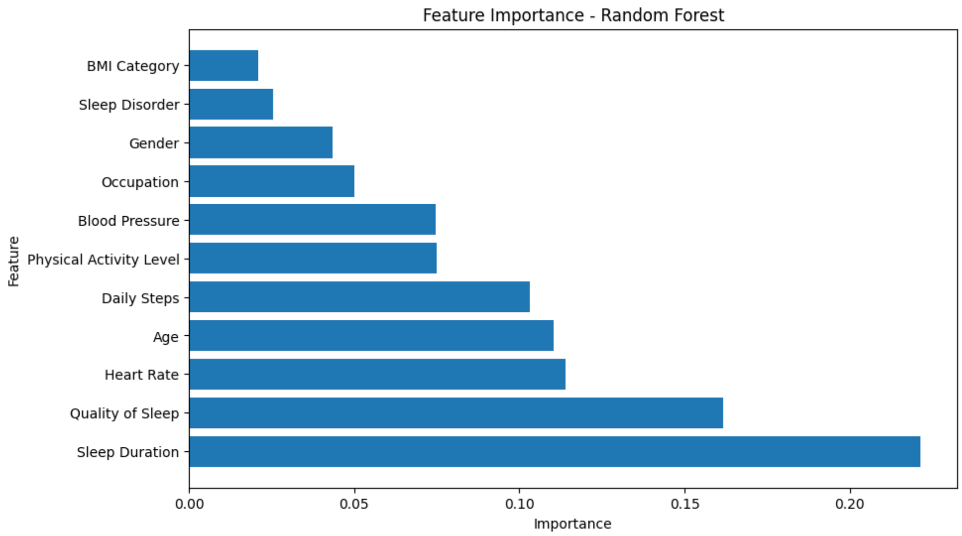
5. Sleep-related variables contributed substantially to model accuracy.

Hence, the machine learning experiments demonstrated that stress levels can be predicted with high accuracy using sleep, lifestyle, and cardiovascular indicators. Among the evaluated algorithms, Random Forest produced the best performance, indicating its suitability for healthcare-related predictive analytics tasks.

Feature Importance Analysis:

To identify the factors that contributed most significantly to stress level prediction, a feature importance analysis was performed using the Random Forest model. Feature importance measures the relative contribution of each predictor variable to the model's decision-making process. Higher importance values indicate a greater influence on the prediction outcome.

Feature	Importance
Sleep Duration	0.221510
Quality of Sleep	0.161708
Heart Rate	0.113826
Age	0.110289
Daily Steps	0.103252
Physical Activity Level	0.074992
Blood Pressure	0.074787
Occupation	0.049885
Gender	0.043363
Sleep Disorder	0.025504
BMI Category	0.020885



A feature importance plot was generated to visually compare the contribution of each variable.

Observation:

Sleep Duration emerged as the most influential predictor with an importance score of 0.2215. Quality of Sleep was identified as the second most important feature, followed by Heart Rate, Age, and Daily Steps.

Interestingly, BMI Category and Sleep Disorder exhibited relatively low importance scores, suggesting that these variables contributed less to stress level prediction within the dataset.

Interpretation:

Earlier correlation analysis indicated that Quality of Sleep had the strongest relationship with Stress Level (-0.90). However, Random Forest feature importance analysis revealed that Sleep Duration contributed more significantly to the model's predictions.

This observation highlights an important distinction between correlation analysis and feature importance. Correlation measures the strength of a direct relationship between two variables, whereas feature importance evaluates the contribution of a variable when all predictors are considered simultaneously.

The results suggest that although Quality of Sleep is strongly associated with Stress Level, Sleep Duration provides the most valuable predictive information when combined with other health indicators.

Conclusion:

The feature importance analysis demonstrated that sleep-related variables are the dominant predictors of stress levels. Sleep Duration and Quality of Sleep collectively contributed the most to model performance, reinforcing the significance of healthy sleep patterns in stress management and prediction.

10. Limitations and Future Scope

Limitations

1. The dataset contains only 374 records, which limits the generalizability of the findings.
2. The dataset is synthetic and may not fully represent real-world healthcare populations.
3. Stress levels are influenced by numerous psychological, social, and environmental factors that are not included in the dataset.
4. The study relied on a single dataset and therefore may not capture population-specific variations.

Future Scope

1. Future studies may utilize larger real-world healthcare datasets.
2. Additional variables such as mental health indicators, dietary habits, and wearable sensor data can be incorporated.
3. Advanced machine learning and deep learning techniques can be explored for improved prediction performance.
4. The developed model can be integrated into healthcare monitoring systems for early stress assessment and preventive intervention.

Conclusion:

This study investigated the relationship between sleep health, lifestyle factors, cardiovascular indicators, and stress levels using machine learning techniques.

Exploratory data analysis revealed strong associations between sleep-related variables and stress levels. Correlation analysis indicated that Quality of Sleep and Sleep Duration were strongly inversely related to Stress Level.

Three machine learning algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were implemented and evaluated. Among these models, Random Forest achieved the highest prediction accuracy of 100%, demonstrating its effectiveness in capturing complex relationships within the dataset.

Feature importance analysis further revealed that Sleep Duration was the most influential predictor, followed by Quality of Sleep and Heart Rate. These findings emphasize the critical role of healthy sleep patterns in stress management.

Overall, the results demonstrate the potential of machine learning-based healthcare analytics for stress prediction and highlight the importance of sleep-related factors in understanding individual well-being.

REFERENCES

1. Kaggle – for the dataset
Link: https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset

2.ChatGpt(OpenAI)-for code and report Assistance:
Link: https://www.chatgpt.com